RetailPulse: Customer Retention & Revenue Intelligence System

End-to-End Data Analytics Project

A Data-Driven Approach to Retail Growth & Customer Retention

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Tools Used:
Excel • MySQL • Python • Power BI • GitHub

Dataset: UCI Online Retail Dataset

Time Period: Dec 2010 – Dec 2011

Date: 2026

Executive Summary

RetailPulse is an end-to-end data analytics project designed to analyze retail transaction data and generate actionable business insights.

The project analyzes 392,692 transactions across 4,338 customers and 37 countries.

Key Highlights:

- Total Revenue: £8.89M
- Total Orders: 19K
- Avg Order Value: £479.56
- Peak Month: November (£1.16M)
- UK contributes 82% of revenue

Objective:

To understand revenue trends, customer behavior, and identify growth opportunities.

This project demonstrates how data analytics can drive business decisions and improve revenue and retention strategies.

Business Problem

The business lacks visibility into:

- Revenue trends
- Customer retention
- Product performance
- Geographic contribution

Objectives

- Analyze monthly revenue trends
- Identify top products and customers
- Perform customer segmentation (RFM)
- Provide actionable business insights

Dataset Overview

Attribute	Value
Source	UCI Online Retail
Period	Dec 2010 – Dec 2011
Transactions	392,692
Customers	4,338
Countries	37

Columns

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country
- Revenue

Data Cleaning

Data cleaning ensured accuracy and reliability for further analysis.

Performed using Excel:

- Removed null CustomerID
- Removed negative quantities
- Removed duplicates
- Standardized date format
- Created Revenue column

Result:

- Clean dataset: 392,692 rows
- No missing or invalid values

SQL Analysis

SQL analysis helped extract key business metrics and identify revenue patterns.

Performed analysis using SQL queries:

- Total revenue calculation
- Monthly revenue trend
- Top products by revenue
- Top customers
- Revenue by country
- Peak shopping hours

Key Result:

November recorded highest revenue (£1.16M)

Python Analysis

Tools used:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Analysis Performed:

- Revenue trend visualization
- Correlation analysis
- Customer behavior analysis

Key Finding:

Revenue is strongly correlated with quantity (bulk purchasing behavior).

Customer Segmentation (RFM)

RFM segmentation helped classify customers based on behavior for targeted marketing strategies.

RFM stands for:

- Recency (last purchase)
- Frequency (number of purchases)
- Monetary (total spend)

Segments:

- Champions
- Loyal Customers
- At Risk
- Lost Customers

Key Insight:

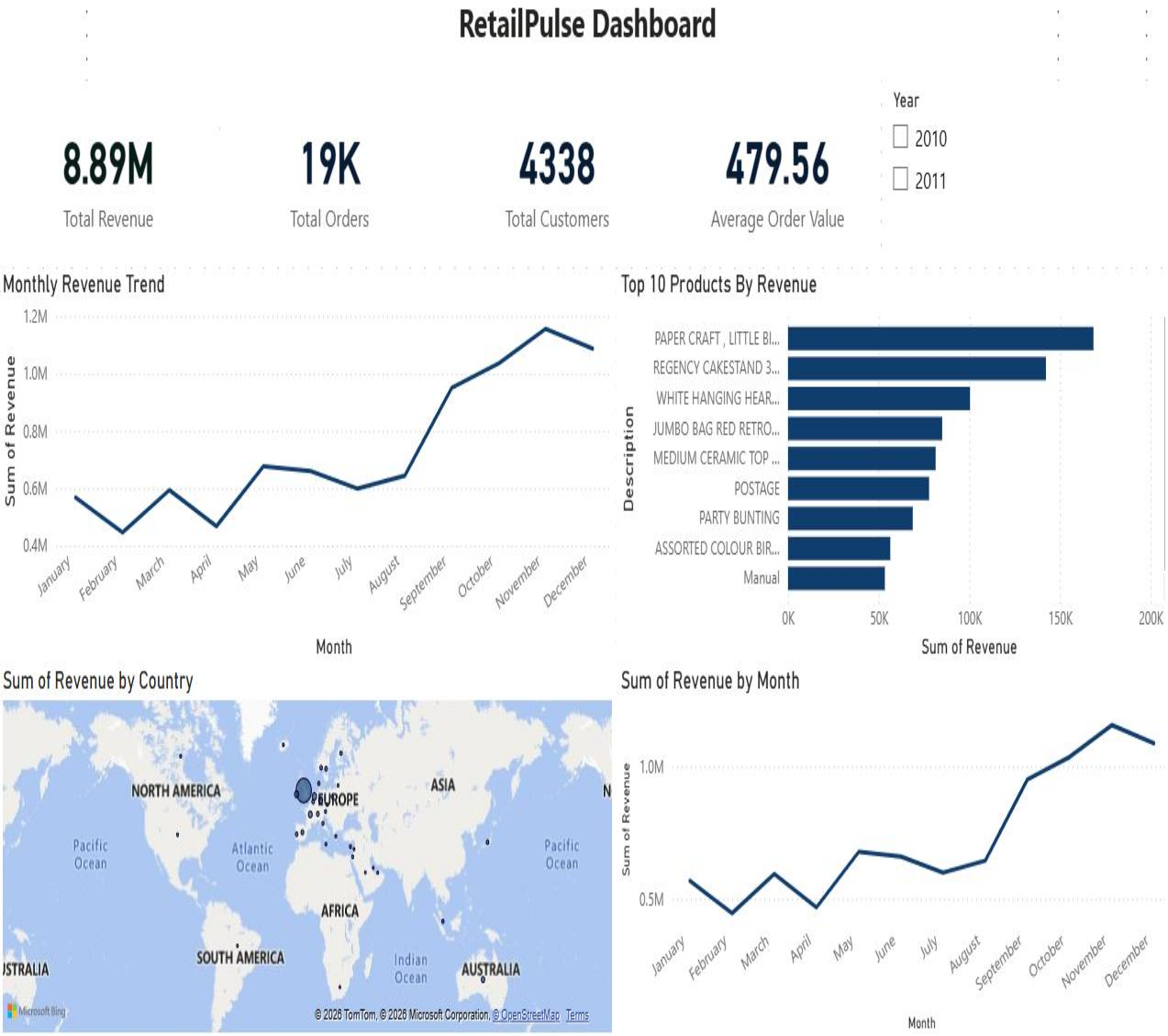
Most customers fall under At Risk / Lost segments, indicating low retention.

Power BI Dashboard

The dashboard provides an interactive interface to monitor KPIs, analyze trends, and explore data across different dimensions such as time and geography.

Dashboard includes:

- KPI Cards (Revenue, Orders, Customers)
- Monthly Revenue Trend
- Top 10 Products
- Revenue by Country (Map)
- Year Filter (Slicer)



Key Insights

Revenue

- Peak sales in November
- Strong Q4 growth

Geography

- UK dominates revenue (82%)
- Opportunity in Europe

Customers

- Few customers generate majority revenue
- Retention is weak

Products

- Top products contribute most revenue

A small percentage of products and customers contribute to a large portion of revenue (Pareto principle).

Business Recommendations

- Expand into international markets
- Improve customer retention strategies
- Focus on top-performing products
- Increase marketing during low months

Conclusion

RetailPulse demonstrates the complete data analytics lifecycle, transforming raw data into actionable insights. The project highlights how businesses can leverage data to improve revenue, enhance customer retention, and make strategic decisions.The insights generated can help improve:

- Revenue growth
- Customer retention
- Market expansion

Tools & Technologies

- Excel → Data Cleaning
- MySQL → Data Analysis
- Python → EDA & Segmentation
- Power BI → Dashboard